CS 473 (Spring 2025) Yiming Qin (yimingq3@illinois.edu)

Problem 8.3

Solution:

Let

$$\mathcal{S} \;=\; \Big\{\, S \subseteq \{1,\dots,n\} : 1 \le |S| \le 3,\ \sum_{i\in S} p_i \le P \Big\}$$

be the set of all possible combinations of tasks of size at most 3 whose total power draw is at most P. $|\mathcal{S}| = O(n^3)$.

For each combination $S \in \mathcal{S}$, introduce a variable x_S representing the total number of running hours of S.

Task i must be run for a total of h_i hours (can be split across many states). Whenever we are in state S and $i \in S$, we accrue work toward task i. Hence we enforce

$$\sum_{\substack{S\in\mathcal{S}\\ i\in S}} x_S = h_i, \quad i = 1,\dots,n.$$

$$x_S \;\ge\; 0, \quad \forall S \in \mathcal{S}.$$

Minimize the total time used:

$$\min \sum_{S\in\mathcal{S}} x_S.$$

Putting it together, we obtain the LP problem:

$$\begin{aligned}
\text{Minimize} \quad & \sum_{S\in\mathcal{S}} x_S \\
\text{s.t.} \quad & \sum_{\substack{S\in\mathcal{S}\\ i\in S}} x_S \;=\; h_i, \quad i = 1,\dots,n, \\
& x_S \;\ge\; 0, \qquad\quad \forall\, S \in \mathcal{S}.
\end{aligned}$$

Algorithm:

1. Construct and solve the LP to get optimal values $\{x_S^*\}_{S\in\mathcal{S}}$.
2. List all states S with $x_S^* > 0$.
3. Build the schedule by concatenating, in any order, one time block of length x_S^* during which we run the tasks in S. Because tasks are fully preemptible, we can schedule these blocks arbitrarily.

The resulting schedule has total duration $\sum_S x_S^*$, meets the "≤ 3 tasks" and "$\leq P$ power" limits by the construction above, and gives each task i exactly $\sum_{S \ni i} x_S^* = h_i$ hours of run-time. There are $|\mathcal{S}| = O(n^3)$ variables and $O(n)$ constraints, both are polynomial in n.